

4.1.1 – 4.1.2 Data Sources and Database Connection

خمسہ اجزاء: ڈیٹا بیس، ڈرائیور، کنکشن، کرسر، ایس کیو ایل۔

CS XII

Chapter 4: Data and Analysis

Lecture 02 of 8

★ Golden

4.1.1 – 4.1.2 Data Sources and Database Connection ★ Golden

LEARN IT

Data sources are where information comes from: school records, websites, sensors, databases. **Connectivity** is the conversation between your Python program and that source so you can read and update data.

Five components of a database connection, in the order they actually work:

#	Component	Role
1	Database	Organised store (SQLite = one .db file; MySQL = a server for big apps)
2	Driver	Translator between Python and that DBMS (<code>sqlite3</code> is built in)
3	Connection object	Opens, manages and closes the pipeline
4	Cursor object	Runs SQL and walks through result rows
5	SQL queries	CREATE, INSERT, SELECT, UPDATE, DELETE



EXAM TIP

Write the five names in order and one line each. A common trap is swapping cursor and connection: the connection is the pipe; the cursor is the messenger that travels through it.

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CHECK YOURSELF

1. Which object executes `SELECT *` ?
2. Is SQLite a driver, a database, or both?

Answers: 1. The cursor. 2. SQLite is the database; the `sqlite3` module is the driver.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The five connection parts in order start with:

- a. Database then driver
- b. Cursor then database
- c. SQL then mouse
- d. Chart then file

MCQ 2. The cursor:

- a. Executes SQL and fetches rows
- b. Is the .db file
- c. Is matplotlib
- d. Is Figma

MCQ 3. The connection object is the:

- a. Pipeline
- b. SQL text
- c. Driver brand only
- d. Pie

MCQ 4. sqlite3 is the:

- a. Built-in driver (+ DB engine in-process)
- b. A web browser
- c. An OSI layer
- d. A sort

MCQ 5. SQL examples:

- a. SELECT, INSERT, CREATE
- b. def, for, if
- c. AND, OR, NOT gates
- d. HTTP GET only

MCQ 6. Swapping cursor and connection in an answer is a:

- a. Classic 0-mark mix-up
- b. Required trick
- c. Big O class
- d. UX heuristic

Extra short questions

1. Name the five components in order.
2. Why a driver?
3. SQLite vs MySQL in one line.
4. Draw the flow.

Extra long question

1. Label a 9-line sqlite snippet with the five components. Explain commit vs close. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Database, driver, connection, cursor, SQL.
2. Translates Python calls to the DBMS protocol/API.
3. SQLite: file, no server. MySQL: server for many users.
4. Python → driver → connection → cursor → SQL → DB.

Long

1. import=driver; connect=connection; cursor=cursor; execute=SQL;
school.db=database. commit saves; close unlocks. Missing commit → empty next run.

Next lecture: 4.2 Creating a SQLite Database in Python